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Inventor(s)

JAIN; Ankit et al.

SYSTEM AND A METHOD FOR MANAGING VR DEVICE OPERATIONS IN AN IoT ENVIRONMENT

Abstract

A method for a Virtual Reality (VR) device in an Internet of Things (IoT) environment, the method may include: obtaining sensor data from the VR device; identifying at least one discontinuity in at least one operation associated with the VR device while a user is using the VR device, based on the sensor data; determining at least one ambient parameter around the VR device causing the at least one discontinuity in the at least one operation; identifying at least one IoT device capable of modifying the at least one ambient parameter; and controlling one or more operational characteristics of the at least one IoT device to modify the at least one ambient parameter around the VR device to reduce the at least one discontinuity in the at least one operation.

Inventors: JAIN; Ankit (Delhi, IN), Samal; Siba Prasad (Bangalore, IN)

Applicant: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

Family ID: 1000008111096

Assignee: SAMSUNG ELECTRONICS CO., LTD. (Suwon-si, KR)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation application, claiming priority under § 365 (c), of an International application No.

PCT/KR2024/009511, filed on Jul. 4, 2024, which is based on and claims the benefit of a Indian Patent Application number 202441013757, filed on Feb. 26, 2024, in the Indian Patent Office, the disclosure of which is incorporated by reference herein in its entirety.

BACKGROUND

1. Field

[0002] The present disclosure, in general, relates to systems and methods for managing virtual reality (VR) device operations in an Internet of Things (IoT) environment. More particularly, the disclosure relates to identifying discontinuities in the operation of the VR device due to an ambient parameter associated with an ambient environment and modifying the ambient parameter by using a nearby IoT device to reduce the discontinuities.

2. Description of Related Art

[0003] As technology continues to advance, virtual reality (VR) devices have emerged as innovative devices that provide users with immersive experiences. However, the optimal functioning of VR devices like VR headsets heavily depends on the performance of their sensors under specific operating conditions. When these specific operating conditions are not met, it can lead to suboptimal performance, causing various issues that can disrupt the user experience.

[0004] For example, the sensors may irregularly start and stop. In particular, the sensors may start providing sensor values when optimal ambient conditions are met and may irregularly stop when the optimal ambient conditions slightly vary, despite the user being engaged with the VR device. For example, consider a scenario in which the user is wearing the VR headset and due to the user's movement in the direction of a light source, the ambient illumination value suddenly shoots up. Consider another example scenario in which the user goes into a dark-lit room, and hence the ambient illumination value goes to zero. Thus, ambient parameters like the ambient illumination value have caused a discontinuity in the operation of the VR device.

[0005] Further, the sensors may provide inconsistent sensor values or sensor readings. The inconsistent sensor values can result in periods where the sensor readings become steady, hindering accurate user interaction and experience. This generally happens when the sensor is unable to fetch the latest sensor values and hence provides the previously fetched sensor value instead. For example, consider a scenario where for hand tracking, the sensor might not be able to detect a new position and so gives the last position coordinates. Thus, due to the inconsistent performance of the sensor, there is a discontinuity in the operation of the VR device.

[0006] Additionally, the sensors may provide non-co-relatable sensor values which can occur when multiple sensors are triggered simultaneously, leading to discrepancies and challenges in interpreting user actions and environmental factors accurately. For example, due to the fog on the display, the motion tracking may give new motion coordinates but while performing an eye-based navigation tracking, the VR device is unable to give coordinates for the same.

[0007] When the ambient parameter causes discontinuity in the performance of VR headsets, the users are required to manually adjust the ambient environment without guidance from the headset

itself. This manual intervention disrupts the immersive experience and adds an extra layer of complexity for users. Rather than seamlessly adapting to changes in ambient conditions, the user must take the initiative to understand and modify the ambient parameters so that the VR headset can function continuously. This process can be cumbersome and detracts from the immersive and user-friendly nature of VR technology.

[0008] Thus, the proper functioning of the VR headsets is linked to the performance of their sensors, which are sensitive to specific operating conditions in the ambient environment.

Irregularities in the sensor behaviour, discontinuities in the performance of the sensor due to the changes in the ambient conditions in the ambient environment, and the lack of solutions for improving the ambient parameters by the headset underline the challenges faced by the users in maintaining the immersive and seamless virtual experience.

SUMMARY

[0009] This summary is provided to introduce a selection of concepts, in a simplified format, that are further described in the detailed description. This summary is neither intended to identify key or essential inventive concepts of the disclosure and nor is it intended for determining the scope of the disclosure.

[0010] According to an embodiment of the present disclosure, a method for a Virtual Reality (VR) device in an Internet of Things (IoT) environment, the method may include obtaining sensor data from the VR device. According to an embodiment of the present disclosure, a method for a Virtual Reality (VR) device in an Internet of Things (IoT) environment, the method may include identifying at least one discontinuity in at least one operation associated with the VR device while a user is using the VR device, based on the sensor data. According to an embodiment of the present disclosure, a method for a Virtual Reality (VR) device in an Internet of Things (IoT) environment, the method may include determining at least one ambient parameter around the VR device associated with the at least one discontinuity in the at least one operation. According to an embodiment of the present disclosure, a method for a Virtual Reality (VR) device in an Internet of Things (IoT) environment, the method may include identifying at least one IoT device capable of modifying the at least one ambient parameter. According to an embodiment of the present disclosure, a method for a Virtual Reality (VR) device in an Internet of Things (IoT) environment, the method may include controlling one or more operational characteristics of the at least one IoT device to modify the at least one ambient parameter around the VR device to reduce the at least one discontinuity in the at least one operation.

[0011] According to an embodiment of the present disclosure, an apparatus configured to connect with a Virtual Reality (VR) device in an Internet of Things (IoT) environment, the apparatus may include at least one memory storing instructions. According to an embodiment of the present disclosure, an apparatus configured to connect with a Virtual Reality (VR) device in an Internet of Things (IoT) environment, the apparatus may include one or more processors operatively connected to the at least one memory. According to an embodiment of the present disclosure, one or more processors may be configured to obtain sensor data from the VR device. According to an embodiment of the present disclosure, one or more processors may be configured to identify at least one discontinuity in at least one operation associated with the VR device while a user is using the VR device, based on the sensor data. According to an embodiment of the present disclosure, one or more processors may be configured to determine at least one ambient parameter around the VR device associated with the at least one discontinuity in the at least one operation. According to an embodiment of the present disclosure, one or more processors may be configured to identify at least one IoT device capable of modifying the at least one ambient parameter. According to an embodiment of the present disclosure, one or more processors may be configured to control one or more operational characteristics of the at least one IoT device to modify the at least one ambient parameter around the VR device to reduce the at least one discontinuity in the at least one operation.

[0012] One embodiment provides a machine readable medium containing instructions. The instructions, when executed by at least one processor, may cause the at least one processor to perform the method corresponding.

[0013] To further clarify the advantages and features of the disclosure, a more particular description of the disclosure will be rendered by reference to specific embodiments thereof, which is illustrated in the appended drawings. It is appreciated that these drawings depict only typical embodiments of the disclosure and are therefore not to be considered limiting of its scope. The disclosure will be described and explained with additional specificity and detail with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0014] These and other features, aspects, and advantages of the disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

[0015] FIG. 1 illustrates an example of IoT environment having a smart device to manage HMD device operations, according to an embodiment of the present disclosure;

[0016] FIG. 2 illustrates a detailed block diagram of a system for managing VR device operations in the IoT environment, according to an embodiment of the present disclosure;

[0017] FIG. 3 illustrates various components included in the HMD device and the smart device respectively, according to an embodiment of the present disclosure;

[0018] FIG. 4 illustrates an overview of the working of the discontinuity detector module, according to an embodiment of the present disclosure;

[0019] FIG. 5 illustrates a method for identifying a discontinuity in the operations associated with the HMD device, according to an embodiment of the present disclosure;

[0020] FIG. 6 illustrates a method for the determination of the ambient parameters by the discontinuity correlator module, according to an embodiment of the present disclosure;

[0021] FIG. 7 illustrates a method implemented in the capability correlator module, according to an embodiment of the present disclosure;

[0022] FIG. 8 illustrates a method flow of the smart device, according to an embodiment of the present disclosure; and

[0023] FIG. 9 illustrates an example scenario when the user is using an XR headset in a dimly lit environment.

[0024] Further, skilled artisans will appreciate that elements in the drawings are illustrated for simplicity and may not have been necessarily been drawn to scale. For example, the flow charts illustrate the method in terms of the most prominent operations involved to help to improve understanding of aspects of the disclosure. Furthermore, in terms of the construction of the device, one or more components of the device may have been represented in the drawings by conventional symbols, and the drawings may show only those specific details that are pertinent to understanding the embodiments of the disclosure so as not to obscure the drawings with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

DETAILED DESCRIPTION

[0025] For the purpose of promoting an understanding of the principles of the disclosure, reference will now be made to the embodiment illustrated in the drawings and specific language will be used to describe the same. It will nevertheless be understood that no limitation of the scope of the disclosure is thereby intended, such alterations and further modifications in the illustrated system, and such further applications of the principles of the disclosure as illustrated therein being contemplated as would normally occur to one skilled in the art to which the disclosure relates.

[0026] It will be understood by those skilled in the art that the foregoing general description and

the following detailed description are explanatory of the disclosure and are not intended to be restrictive thereof.

[0027] Reference throughout this specification to “an aspect”, “another aspect” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment of the disclosure. Thus, appearances of the phrase “in an embodiment”, “in another embodiment” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment.

[0028] The terms “comprises”, “comprising”, or any other variations thereof, are intended to cover a non-exclusive inclusion, such that a process or method that comprises a list of operations does not include only those operations but may include other operations not expressly listed or inherent to such process or method. Similarly, one or more devices or sub-systems or elements or structures or components preceded by “comprises . . . a” does not, without more constraints, preclude the existence of other devices or other sub-systems or other elements or other structures or other components or additional devices or additional sub-systems or additional elements or additional structures or additional components.

[0029] FIG. 1 illustrates an example of IoT environment having a smart device to manage HMD device operations, according to an embodiment of the present disclosure. According to an embodiment, the IoT environment includes a system **100**. In an embodiment, the system **100** may include a Head Mounted Device (HMD) device **101**, a smart device **103**, and one or more IoT devices **105**. Consider a scenario where the HMD device **101** is operatively coupled with the smart device **103** via a communication network **107**. According to an embodiment, the smart device **103** may be further operatively coupled with the one or more IoT devices **105** via the communication network **107**. In an example, scenario, the one or more IoT devices may include but are not limited to, a smart light bulb **105-1**, a smart fan **105-2**, a smart washing machine **105-3**, a smart chimney **105-4**, a smart air conditioner (AC) **105-5**, and the like. It is to be noted that one or more IoT devices are collectively referred to as one or more IoT devices **105**. Further, the one or more IoT devices **105** may be interchangeably referred to as IoT devices **105** throughout the disclosure.

[0030] In an example, the HMD device **101** may be implemented with technologies like Virtual Reality (VR), extended reality (XR), Augmented Reality (AR), and Mixed Reality (MR). Accordingly, the HMD device **101** may be alternatively referred to as a VR device, an XR device, an AR device, and an MR device depending on the specific technology integrated within the HMD device **101**. In an example, the smart device **103** may correspond to a hub device such as a smart central home monitoring device, smart television, family hub, tablet, personal computer (PC), monitor, hub core software, and the like. In an example, the IoT devices **105** may correspond to a smart light bulb **105-1**, a smart fan **105-2**, a smart washing machine **105-3**, a smart chimney **105-4**, a smart air conditioner (AC) **105-5**, and the like.

[0031] Consider an example scenario, where a user is using the HMD device **101** in the IoT environment without an active airflow. The lack of active airflow may occur due to different swing levels, fan speed, swing direction of the smart AC **105-5** or the smart fan **105-2**, and the like. As a result, the lenses of the headset may begin to fog up while the user is using the HMD device **101**, leading to diminished display visibility to the user. Thus, the discontinuity occurred in the operation of the HMD device **101**. This adversely affects the operation of the headset and further leads to a bad user experience. In an example, the discontinuity may be abnormally functioning of HMD headset as the hand tracking and the headset tracking features are not working properly in changing environment (e.g., brightness, humidity, etc.).

[0032] According to an embodiment, the system **100** may overcome the various issues as discussed in the above paragraphs. In an embodiment, the system **100** may identify at least one discontinuity in at least one operation associated with the HMD device **101** while a user is using the HMD device. The system **100** further correlates the discontinuity with ambient parameters around the HMD device **101** to determine the IoT devices and capabilities of the IoT devices that can be

modified to reduce the discontinuities in at least one operation associated with the HMD device **101**. A detailed operation of the system **100** will be explained in the forthcoming paragraphs. [0033] FIG. **2** illustrates a detailed block diagram of the system **100** for managing VR device operations in the IoT environment, according to an embodiment of the present disclosure. In an embodiment, the HMD device **101** may include an HMD processor(s) **201**, an HMD database **203**, an HMD transceiver **205**, an HMD memory **207**, an HMD Module(s) **209**, an HMD sensor unit **211**, and an HMD AI module **213**. Likewise, the smart device **103** may include a processor(s) **215**, a database **217**, a transceiver **219**, a memory **221**, a module(s) **223**, and an AI module **225**. Further, each IoT device **105** may include an IoT processor(s) **227**, an IoT database **229**, an IoT transceiver **231**, an IoT memory **233**, an IoT module(s) **235**, and an IoT AI module **237**. The above memories may each be more than one memory.

[0034] In an example, the HMD processor(s) **201**, the processor(s) **215**, and the IoT processor(s) **227** may be operatively coupled with their respective module(s) i.e. the HMD module(s) **209**, the module(s) **223**, and the IoT module(s) **235** respectively for processing, executing or performing a set of operations. In an embodiment, the HMD processor(s) **201**, the processor(s) **215**, and the IoT processor(s) **227** may include at least one data processor for executing processes in a Virtual Storage Area Network. The HMD processor(s) **201**, the processor(s) **215**, and the IoT processor(s) **227** may include specialized processing units such as integrated system (bus) controllers, memory management control units, floating point units, graphics processing units, digital signal processing units, etc. In one embodiment, the HMD processor(s) **201**, the processor(s) **215**, and the IoT processor(s) **227** may include a central processing unit (CPU), a graphics processing unit (GPU), or both. The HMD processor(s) **201**, the processor(s) **215**, and the IoT processor(s) **227** may be one or more general processors, digital signal processors, application-specific integrated circuits, field-programmable gate arrays, servers, networks, digital circuits, analog circuits, combinations thereof, or other currently-known or later developed devices for analyzing and processing data. The HMD processor(s) **201**, the processor(s) **215**, and the IoT processor(s) **227** may execute one or more instructions, such as code generated manually (i.e., programmed) to perform one or more operations disclosed herein throughout the disclosure.

[0035] In an example, the HMD database **203**, the database **217**, and the IoT database **229** may be implemented with integrated hardware and software. The hardware may include a hardware disk controller with programmable search capabilities or a software system running on general-purpose hardware. Examples of the HMD database **203**, the database **217**, and the IoT database **229** are but are not limited to, in-memory databases, cloud databases, distributed databases, embedded databases, and the like. The HMD database **203**, the database **217**, and the IoT database **229**, among other things, serve as a repository for storing data processed, received, and generated by one or more of the HMD processor(s) **201**, the processor(s) **215**, and the IoT processor(s) **227**, and the HMD module(s) **209**, the module(s) **223**, and the IoT module(s) **235**.

[0036] In an example, the HMD transceiver **205**, the transceiver **219**, and the IoT transceiver **231** may be a transmitter and a receiver unit. The HMD transceiver **205**, the transceiver **219**, and the IoT transceiver **231** may communicate with each other via the communication network implemented with any of the wireless standards, such as 3G, 4G, 5G, or the like. Further, other wireless techniques, such as Wi-Fi, Bluetooth, etc. may also be used by the HMD transceiver **205**, the transceiver **219**, and the IoT transceiver **231** for communicating with each other.

[0037] In an example, the HMD memory **207**, the memory **221**, and the IoT memory **233** may include any non-transitory computer-readable medium known in the art including, for example, volatile memory, such as static random-access memory (SRAM) and dynamic random-access memory (DRAM), and/or non-volatile memory, such as read-only memory (ROM), erasable programmable ROM, flash memories, hard disks, optical disks, and magnetic tapes. Each of the HMD memory **207**, the memory **221**, and the IoT memory **233** are communicatively coupled with their respective processors i.e. the HMD processor(s) **201**, the processor(s) **215**, and the IoT

processor(s) **227** to store user usage data, sensor data (e.g. past sensor values collected by the sensors at previous time instances, current sensors values collected by the sensors at current time instances, continuous state of the sensor data, pre-stored sensor data and the like) or processing instructions for completing one or more processes. Furthermore, each of the HMD memory **207**, the memory **221**, and the IoT memory **233** may include their respective the HMD database **203**, the database **217**, and the IoT database **229** to store the information as required by their respective HMD module(s) **209**, the module(s) **223**, and the IoT module(s) **235** and the HMD processor(s) **201**, the processor(s) **215**, and the IoT processor(s) **227**. Further, the HMD memory **207**, the memory **221**, and the IoT memory **233** may store one or more values, such as, but not limited to, one or more output values of the sensor data, threshold values, etc. Furthermore, the HMD memory **207**, the memory **221**, and the IoT memory **233** may store the HMD module(s) **209**, the module(s) **223**, IoT module(s) **235**, or one or more models for performing operations as disclosed throughout the disclosure.

[0038] As an example, the HMD module(s) **209**, the module(s) **223**, and the IoT module(s) **235** may include a program, a subroutine, a portion of a program, a software component, or a hardware component capable of performing a stated task or function. As used herein, the HMD module(s) **209**, the module(s) **223**, and the IoT module(s) **235** may be implemented on a hardware component such as a server independently of other modules, or a module can exist with other modules on the same server, or within the same program, or in the cloud, Smart-home edge software, or on the IoT devices **105**. The HMD module(s) **209**, the module(s) **223**, and the IoT module(s) **235** may be implemented on a hardware component such as processor one or more microprocessors, microcomputers, microcontrollers, digital signal processors, central processing units, state machines, logic circuitries, and/or any devices that manipulate signals based on operational instructions. The HMD module(s) **209**, the module(s) **223**, and the IoT module(s) **235** when executed by the HMD processor(s) **201**, the processor(s) **215**, and the IoT processor(s) **227** may be configured to perform any of the described functionalities.

[0039] As an example, the HMD sensor unit **211** may include a plurality of sensors, tracking cameras, microphones, scanners, Light Emitting Diode (LED) illuminators, IR cameras, a lip tracking camera, IR Illuminators, 6DOF Cameras, SLAM Cameras, Face tracking Camera, and the like. In an example, the plurality of sensors includes camera sensors, motion sensors, depth sensors, heart rate sensors, proximity sensors, tracking sensors, sound sensors, gyro sensors, acceleration sensors, pressure sensors, and hall effect sensors. In an embodiment, the HMD sensor unit **211** may send the sensor data to the smart device **103** for identifying at least one discontinuity in the at least one operation associated with the VR device while a user is using the VR device.

[0040] In an example, the HMD AI module **213**, the AI module **225**, and the IoT AI module **237** may include a plurality of neural network layers. Examples of neural networks include, but are not limited to, convolutional neural network (CNN), deep neural network (DNN), recurrent neural network (RNN), and Restricted Boltzmann Machine (RBM). The learning technique is a method for training a predetermined target device using a plurality of learning data to cause, allow, or control the target device to make a determination or prediction. Examples of learning techniques include, but are not limited to, supervised learning, unsupervised learning, semi-supervised learning, or reinforcement learning. At least one of a plurality of CNN, DNN, RNN, RBM models and the like may be implemented to thereby achieve execution of the present subject matter's mechanism through an AI model. A function associated with AI may be performed through the non-volatile memory, the volatile memory, and the processor. The processor may include one or a plurality of processors. At this time, one or a plurality of processors may be a general purpose processor, such as a central processing unit (CPU), an application processor (AP), or the like, a graphics-only processing unit such as a graphics processing unit (GPU), a visual processing unit (VPU), and/or an AI-dedicated processor such as a neural processing unit (NPU). The one or more plurality of processors control the processing of the input data in accordance with a predefined

operating rule or artificial intelligence (AI) model stored in the non-volatile memory and the volatile memory. The predefined operating rule or artificial intelligence model is provided through training or learning.

[0041] FIG. 3 illustrates various components included in the HMD device **101** and the smart device **103** respectively, according to an embodiment of the present disclosure. According to an embodiment, the HMD module(s) **209** of the HMD device **101** or the module(s) **223** of the smart device **103** may include a user usage module **301**, a discontinuity detector module **303**, a discontinuity correlator module **305**, a capability correlator module **307**, and the operational characteristic fixer module **309**. According to an embodiment, the user usage module **301**, the discontinuity detector module **303**, and the discontinuity correlator module **305** may be implemented in the HMD device **101** and the capability correlator module **307**, and the operational characteristic fixer module **309** may be implemented in the smart device **103**. According to some embodiment, the processors i.e. the HMD processor **201**, the processor **215**, associated with the respective modules may cause the aforesaid modules to perform various functions. The detailed working of each component will be explained with respect to the modules in the forthcoming paragraphs for ease of explanation.

[0042] According to an embodiment, the user usage module **301** may get user usage data of the user using the HMD device **101** and the features the user is likely to use. According to an embodiment, as the user starts using the HMD device **101**, the user usage module **301**, at first may identify the user. As an example, the user usage module **301** may identify the user by using credential information provided by the user during logging in the HMD device **101**. Further, the user usage module **301** identifies a current application that the user is running on the HMD device **101** and the at least one operation associated with the HMD device **101**. In an example, the current applications may be related to gaming applications, navigation applications, virtual meetings, and the like. Based on the identification of the current application, the user usage module **301** identifies the at least one operation associated with the HMD device **101**. The at least one operation may be alternately referred to as ‘operations’ hereinafter throughout the disclosure. Accordingly, in an example, the operations associated with the current application may include finger-tracking, audio-tracking, foveated rendering, eye tracking, and the like. According to an embodiment, the user usage module **301** further may identify possible future applications the user may use based on the user history information. An example output of the user usage module **301** is shown in Table 1.

TABLE-US-00001

TABLE 1	Headset Configuration	User	Next Apps	State
Adam	Resident Evil 4: Game	Finger Tracking: On	Passthrough: 0%	Eve Meeting VR: Foveated Rendering: On, Productivity Passthrough: 50%
Elena	Journey of the Gods: 3D Audio: On	Game 3D Touch: On		

[0043] According to an embodiment, the discontinuity detector module **303** may identify at least one discontinuity in the operations associated with the HMD device while the user is using the VR device. FIG. 4 illustrates an overview of the working of the discontinuity detector module **303**, according to an embodiment of the present disclosure. Further, FIG. 5 illustrates a method **500** for identifying the at least one discontinuity in the operations associated with the HMD device, according to an embodiment of the present disclosure. The identification of the at least one discontinuity in the operations will be explained collectively through FIG. 4 and FIG. 5 in the forthcoming paragraphs.

[0044] Referring to FIG. 4 and FIG. 5 the discontinuity detector module **303** inputs **411** from the user usage module **301** and the HMD sensor unit **211**. In an embodiment, the discontinuity detector module **303** may obtain data related to the least one operation associated with the HMD device **101** from the user usage module **301**. For example, consider that in an exemplary embodiment, the user is playing a video game. Further, the identified operation associated with playing the video game may be the hand tracking, the head tracking, a guardian boundary, and the foveated rendering. Further, the discontinuity detector module **303**, at operation **501** of method **500**, may obtain the sensor data associated with the identified operations of the HMD device from the HMD sensor unit

211. For example, the hand tracking often utilizes data from depth-sensing cameras. In an example, the head tracking uses data from the gyroscopes, the accelerometers, and the external cameras. In an example, the guardian boundary may utilize data from the depth cameras or the infrared sensors, and the foveated rendering uses data from the eye-tracking sensors.

[0045] Further, at operation **503**, the discontinuity detector module **303** may correlate the sensor data with the pre-stored sensor data. In an embodiment, the pre-stored sensor data may indicate an optimal working criteria of the operations. In an embodiment, for correlating the sensor data with the pre-stored sensor data, the discontinuity detector module **303** may compare a continuous state of the sensor data with respect to a state of the pre-stored sensor data. Consider an example of assessing the quality of input data streams related to the hand tracking operation. According to an embodiment, the quality assessment of the input data streams is performed for the features and output of the features. In an embodiment, the discontinuity detector module **303** may correlate the inputs and outputs with past data to check if there is a discontinuity. Further, the discontinuity detector module **303** may retrieve the state of the pre-stored sensor data from the HMD database **203** and then compare the continuous state of the sensor data with respect to the pre-stored sensor data. Thus, based on any deviation in the operations of the HMD device **101** as compared to the optimal working criteria of the operations, the discontinuity detector module **303**, at operation **505**, may identify the at least one discontinuity in the operations **413**. The at least one discontinuity indicates a degree of discontinuity in the behaviour of the at least one operation. In an embodiment, the discontinuity detector module **303** further may determine an affecting user state due to the discontinuity, for example navigation loss, or control loss. Table 2 depicts an example of a discontinuity measurement. Further, Table 3 depicts an example of the output provided by the discontinuity detector module **303**. The discontinuity can be an anomaly of the operation of the VR device, or an interruption in sensor data generated by the VR device.

TABLE-US-00002 TABLE 2 Discontinuity Measurement Working Parameters Irregular Tracking: 40% Regular Tracking Irregular Tracking: 70% Regular Tracking against Background

TABLE-US-00003 TABLE 3 Discontinuity Working Discontinuity Operation Measurement Parameters loss effect Hand Irregular Tracking: Regular Tracking Controls Loss Tracking 40% Head Irregular Tracking: Regular Tracking Navigation: Tracking 70% against Background Loss

[0046] According to an embodiment, the discontinuity correlator module **305** may determine ambient parameters around the HMD device **101** causing the at least one discontinuity in the operations. In an embodiment, the discontinuity correlator module **305** may determine the ambient parameters that are causing a problem resulting in the discontinuity of the operations of the HMD device **101**. The discontinuity correlator module **305** takes the output of the discontinuity detector module **303** and provides the ambient parameter that can be modified in order to reduce the discontinuity in the operation of the HMD device **101**. The working of the discontinuity correlator module **305** will be explained in the forthcoming paragraphs.

[0047] FIG. **6** illustrates a method **600** for the determination of the ambient parameters by the discontinuity correlator module **305**, according to an embodiment of the present disclosure. According to an embodiment, at operation **601**, the discontinuity correlator module **305** may obtain the sensor data from the plurality of sensors in the HMD sensor unit **211**, IoT data from the plurality of IoT device **105**, data related to the identified the at least one discontinuity in at least one operation associated with the HMD device **101** and the affecting user state due to the discontinuity. In an example, the IoT data may include the IoT device that was currently operating at the current instance.

[0048] Further, at operation **603**, the discontinuity correlator module **305** may identify at least one of a respective sensor and a respective IoT device that is causing the at least one discontinuity in the operations by using the obtained sensor data and the IoT data. In an example, the discontinuity correlator module **305** may take analog converted raw inputs (i.e. output values of the sensors) that the plurality of sensors are sensing from the IoT devices **105** and the pre-stored sensor data.

Further, the discontinuity correlator module **305** may determine a difference between the output values of the obtained sensor data and the pre-stored sensor data. Based on the determined difference, the discontinuity correlator module **305** may compute a gap in the sensor data with respect to the pre-stored data like no sensor data change, sensor unable to pickup difference, etc. Further, from the non-activation phase of the sensor data, the discontinuity correlator module **305** may determine a type of data that was missed. For example, visual data, audio data, etc. In an embodiment, A correlation may be the computed gap in the sensor data with respect to the pre-stored data like no sensor data change, sensor unable to pickup difference, etc.

[0049] Further, at operation **605**, the discontinuity correlator module **305** may correlate the at least one ambient parameter with the identified at least one discontinuity in the at least one operation. Further at operation **607**, the discontinuity correlator module **305** may determine the at least one ambient parameter that is causing the discontinuity based on the correlation.

[0050] In an embodiment, the discontinuity correlator module **305** further may determine the required modification in the at least one ambient parameter to fill the computed gap to achieve the optimal (or an improved) working of the at least one operation based on the based on the determined difference. Further, the discontinuity correlator module **305** may determine the required modification in the at least one ambient parameter to fill the computed gap to achieve the optimal (or an improved) working of the at least one operation. For example, the discontinuity correlator module **305** may determine what data from the raw data should have been present, but was not present, like “previous object wasn't found” so that when the ambient parameter related to it when modified fills the computed gap. Table 4 illustrates an example output provided by the discontinuity correlator module **305**.

TABLE-US-00004 TABLE 4 Ambient Parameter Needed Change Effecting Parameter Illumination Increase Hand tracking, headset tracking

[0051] As can be seen from the example illustrated in Table 4 the discontinuity correlator module **305** detects the ambient parameter i.e. illumination that is affecting the hand tracking or headset tracking. Thus, if this ambient parameter i.e. illumination is increased then the hand tracking and the headset tracking will be better.

[0052] According to an embodiment, the capability correlator module **307** may determine the IoT devices **105** and the capabilities of the IoT devices. According to an embodiment, the capability correlator module **307** may identify at least one IoT device surrounding the HMD device that is capable of modifying the determined at least one ambient parameter. The detailed working of the capability correlator module **307** and the method for identifying the at least one IoT device surrounding the HMD device that is capable of modifying the determined at least one ambient parameter is explained in the forthcoming paragraphs.

[0053] FIG. 7 illustrates a method implemented in the capability correlator module **307**, according to an embodiment of the present disclosure. According to an embodiment, the capability correlator module **307** may take detected ambient parameters, the required modification in the ambient parameters, and the identified at least one discontinuity in the operations as obtained from the previous modules as an input.

[0054] According to an embodiment, the capability correlator module **307** may determine from a pre-existing map of ambient parameters obtained from the IoT database **229**, the received ambient parameter with the available IoT capabilities with the user at operation **701**. Further, at operation **703**, the capability correlator module **307** may determine the primary capabilities along with the secondary capabilities of the IoT devices **105** that can also cause the discontinuity and have an effect on the ambient parameter. Table 5 depicts an example of the determined primary capabilities and the secondary capabilities of the IoT devices **105**.

TABLE-US-00005 TABLE 5 Primary Capabilities Secondary Capabilities Brightness Light colour, blinds open, display

[0055] Further, at operation **705**, the capability correlator module **307** may determine the location

of the IoT devices that are near the HMD device **101** that have that capability. For example, the IoT device like a smart bulb in the living room where the user is operating with the HMD device **101** is capable of modifying the illumination. Further, at operation **707**, the capability correlator module **307** may check if the IoT devices can be controlled by the user and don't have restrictions. Further, at operation **709**, the capability correlator module **307** may compute a value or possible range of the identified IoT device that is required to be modified so as to match with the ambiance that is required. Table 6 depicts the example of the computed range that is required to be modified.

TABLE-US-00006 TABLE 6 Capabilities IoT device Possible range Light Brightness Smart Light bulb 40%-70% Blinds Open Smart Curtains Open 10% to 40% Display Light Television Screensaver

[0056] Further, at operation **711**, the capability correlator module **307** may obtain all the past routines, etc., that have involved the capability from the IoT database **229**. Table 7 depicts an example of obtained routines.

TABLE-US-00007 TABLE 7 Capabilities IoT devices Routines Light Brightness Light Bulb Power saving routine Blinds Open Smart Curtains Privacy Routine, Night Routine DisplayLight Television Sleeping Routine

[0057] Further, at operation **713**, the capability correlator module **307** may classify the side effects if the ambient parameter related to the capability of the IoT device is modified. Table 8 depicts an example of the classified side effects.

TABLE-US-00008 TABLE 8 Capability Affects Light Brightness Power, Routine met: 40% Blinds Open Routine Violation Display Light No effect

[0058] In an embodiment, after identifying at least one IoT device surrounding the HMD device **101** capable of modifying the determined at least one ambient parameter along with other parameters as explained above, the operational characteristic fixer module **309** may control one or more operational characteristics of the identified at least one IoT device to modify the at least one ambient parameter around the HMD device **101** to reduce the at least one discontinuity in the at least one operation. In an example, the operational characteristic fixer module **309** may perform the required modification based on the capability of the identified at least one IoT device. In an embodiment, the operational characteristic fixer module **309** may perform continuous changes and determine the fixed new values of the IoT devices, and control the settings of the capable IoT devices so that the user can use the HMD device **101** efficiently.

[0059] FIG. **8** illustrates a method flow **800** of the smart device **103**, according to an embodiment of the present disclosure. According to an embodiment, the method **800** is implemented in the smart device **103** via the user usage module **301**, the discontinuity detector module **303**, the discontinuity correlator module **305**, the capability correlator module **307**, the operational characteristic fixer module **309**. The detailed working of each of the modules is explained above, therefore for the sake of brevity, the same is not repeated here.

[0060] According to an embodiment, at operation **801**, the discontinuity detector module **303** may identify the at least one discontinuity in at least one operation associated with the HMD device **101** while the user is using the HMD device **101**. The working of the discontinuity detector module **303** can be envisaged from the explanation provided in FIG. **4**, and FIG. **5**. Further, at operation **803**, the discontinuity correlator module **305** may determine at least one ambient parameter around the HMD device **101** causing the at least one discontinuity in the at least one operation. The working of the discontinuity correlator module **305** can be envisaged from the explanation provided in FIG. **6**. Further at operation **805**, the capability correlator module **307** may identify at least one IoT device surrounding the HMD device **101** capable of modifying the determined at least one ambient parameter. The working of the capability correlator module **307** can be envisaged from the explanation provided in FIG. **7**. Further, at operation **807**, the operational characteristic fixer module **309** may control one or more operational characteristics of the identified at least one IoT device to modify the at least one ambient parameter around the HMD device **101** to reduce the at

least one discontinuity in the at least one operation.

[0061] According to an embodiment, the method **800** may be implemented in the HMD device **101** via the user usage module **301**, the discontinuity detector module **303**, the discontinuity correlator module **305**, the capability correlator module **307**, the operational characteristic fixer module **309**. The explanation of the same is provided through the FIG. 4-FIG. 7. Therefore, for the sake of brevity, the same is not repeated here.

[0062] According to an embodiment, the smart device **103** further may monitor one or more operations of the HMD device when the at least one ambient parameter is modified with respect to the determined required modification, the smart device **103** further may determine, based on the monitoring, the one or more effects on the operation of the VR device due to the modification in the at least one ambient parameter. Further, the smart device **103** may determine a change required in the identified at least one IoT device to fix the determined one or more effects on the operation of the VR device. Further, the smart device **103** notifies the user, the change required in the identified at least one IoT device.

[0063] The forthcoming paragraphs illustrate some use cases to elaborate on the implementation of the disclosed methodology.

[0064] FIG. **9** illustrates an example scenario **900** when the user is using an XR headset in a dimly lit environment **902**. According to an embodiment, due to the dimly lit environment, the headset tracking by the XR headset **901** is not working properly. Due to the low surrounding lights, the hand tracking and the controller tracking are not working properly. According to the implemented method **800**, the user usage module **301** may collect various parameters and track for the XR headset **901**. Further, the discontinuity detector module **303** may identify at least one discontinuity in at least one operation associated with the XR headset **901** while the user is using the VR device. The performance tracked by the user usage module **301** is depicted in Table 9.

TABLE-US-00009 Parameter Performance Tracker Hand tracking 28% Headset Tracking 35% Eye Tracking 32% Pass-through 20% Position tracking 80% Territory marking| Guardian 95% Scaling Lens Fogging 0%

[0065] Based on the tracked performance, the discontinuity detector module **303** tracked the discontinuity and performance degradation in the hand tracking, the headset tracking, and the pass-through visibility. A rule or AI based engine is used to identify the performance degradation or discontinuities. Further, the discontinuity correlator module **305** determines the ambient parameter around the XR device **901** causing the one or more discontinuities and performance degradation. Further, the discontinuity correlator module **305** correlates the at least one ambient parameter with the identified at least one discontinuity. Further, the capability correlator module **307** determines the IoT devices and the capabilities of the IoT devices that can be modified to overcome discontinuities in the affected operation. Table 10 and 11 depict the ambient parameter with the identified at least one discontinuity along with the correlated IoT devices that are capable of modifying the discontinuities in the affected operation.

TABLE-US-00010 TABLE 10 Headset Parameter Environment Parameters Location IoT Devices
Hand Lights Living Blind, Living tracking [intensity/colour/brightness], Room Room Light#1, Natural light Exposure Living Room Light#3 Headset Lights Living Blind, Living Tracking [intensity/colour/brightness] Room Room Light#1, Living Room Light#3 Eye Lights Living Blind, Living Tracking [intensity/colour/brightness] Room Room Light#1, Living Room Light#3 Pass-Lights Living Blind, Living through [intensity/colour/brightness], Room Room Light#1, Object density Living Room Light#3

TABLE-US-00011 TABLE 11 IOT Device Operational Characteristic Change Blind ON, 70% Open Living Room Light#1 Brightness: 90% Colour: Bright White Living Room Light#3 75% Living Room Light#2 Brightness: 70% Colour: off White

[0066] The operational characteristic fixer module **309** modifies the identified IoT devices so that the discontinuities in the performance get reduced. According to the example embodiment,

surrounding lights such as light of smart light bulbs are brightened to fix the issue, and thus the XR device **901** has improved performance, and may perform optimally in improved environment **903**. Table 12 depicts the accuracy obtained in the operations of the XR device **901**.

TABLE-US-00012 TABLE 12 Parameter Working Normalized Percentage Hand tracking 83% Headset Tracking 90% Eye Tracking 75% Pass-through 100%

[0067] Consider a scenario where the user is using a virtual reality fitness platform in “RoomScale” Boundary mode via the HMD device **101**. Thus, it is required to move within a designated space. Further, while using the virtual reality fitness platform the discontinuity detector module **303** has identified a potential issue where the user's guardian boundary gets disrupted, leading to a break in the boundary's continuity. The discontinuity correlator module **305** identified the issue related to the discontinuity, which is a result of the negative space ambient parameter that is crucial for ensuring the seamless operation of the guardian boundary. To address this, the capability correlator module **307** identifies the correlated IoT devices like robot vacuum cleaner (RVC) in the nearby surrounding. Accordingly, the operational characteristic fixer module **309** reconfigures RVC to avoid entering the guardian space, as their presence could obstruct areas of the room and interfere with maintaining the boundary's continuity.

[0068] According to an example scenario, consider that the user is using a fitness application by using the HMD device. The system identified several issues while the user was using the fitness application. For example, due to vigorous exercise, the discontinuity detector module **303** identifies that the user is perspiring heavily, and headset is also slipping. Further, the discontinuity correlator module **305** has identified a disruption in the readings of the heart rate sensor within the HMD. The discontinuity correlator module **305** linked the disruption in the readings of the heart rate sensor to ambient parameters such as air flow and air humidity of the surrounding ambient parameter. In response, the operational characteristic fixer module **309** adjusts the parameters of the nearby IoT devices like the air purifier and the air conditioner to operate in dehumidification mode and decrease the room's temperature, aiming to rectify the discontinuities and effective working of the HMD device.

[0069] According to an example scenario, consider that the user is navigating a virtual environment while using an application to observe real-world spaces. However, the discontinuity detector module **303** detects a disruption in the head tracking functionality when the user moves their head diagonally or alters the yaw and pitch, resulting in the headset's inability to accurately track these movements. Further, the discontinuity correlator module **305** also detects a high illumination in the ambiance. For example, due to the bright light in the ambiance, the system identifies that the user is facing difficulties with head rotation tracking. Further, the discontinuity correlator module **305** identifies that the HMD device is in a low power state and needs to consume additional power to edit the image and mitigate the impact of the excessive light to compensate for excessively high illumination, the HMD device is forced to work harder, utilizing additional computational resources to reduce the effect of the illumination. The capability correlator module **307** attributes this issue to the ambient illumination parameter, deducing that the environment is excessively bright in certain areas. As a solution, the operational characteristic fixer module **309** adjusts the lighting in the room, reducing overall brightness and incorporating more light sources like smart light bulb to ensure continuous and accurate head tracking.

[0070] According to the disclosed techniques, the technical advantage of this system is its real-time environmental adaptation capability. By detecting and analyzing ambient parameters such as illumination, the system can dynamically modify the ambiance parameters to optimize the user experience by reducing the discontinuities. This real-time adaptation enhances immersion and user comfort by addressing issues, ensuring a more seamless and responsive virtual reality experience.

[0071] Unless otherwise defined, all technical and scientific terms used herein have the same meaning as commonly understood by one of ordinary skilled in the art to which this disclosure belongs. The system, methods, and examples provided herein are illustrative only and not intended

to be limiting.

[0072] Benefits, other advantages, and solutions to problems have been described above with regard to specific embodiments. However, the benefits, advantages, solutions to problems, and any component(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential feature or component of any or all the claims.

[0073] While certain embodiments of the disclosure has been particularly shown and described, it will be understood that various changes in form and details may be made therein without departing from the spirit and scope of the following claims.

[0074] According to an embodiment of the disclosure, the method for identifying the at least one discontinuity in the at least one operation may include obtaining the sensor data associated with the at least one operation. According to an embodiment of the disclosure, the method for identifying the at least one discontinuity in the at least one operation may include correlating the sensor data with a pre-stored sensor data indicating an optimal working criteria of the at least one operation to obtain a correlation. According to an embodiment of the disclosure, the method for identifying the at least one discontinuity in the at least one operation may include identifying the at least one discontinuity in the at least one operation based on the correlation.

[0075] According to an embodiment of the disclosure, the correlating the sensor data with the pre-stored sensor data may comprise comparing a continuous state of the sensor data with respect to a state of the pre-stored sensor data.

[0076] According to an embodiment of the disclosure, the at least one discontinuity may indicate a degree of discontinuity in a behaviour of the at least one operation.

[0077] According to an embodiment of the disclosure, the determining the at least one ambient parameter may comprise obtaining, from a plurality of sensors and the at least one IoT device, sensor data and IoT data. According to an embodiment of the disclosure, the determining the at least one ambient parameter may comprise identifying, based on the obtained sensor data and the IoT data, at least one of a respective sensor and a respective IoT device associated with the at least one discontinuity in the at least one operation. According to an embodiment of the disclosure, the determining the at least one ambient parameter may comprise correlating the at least one ambient parameter with the at least one discontinuity in the at least one operation to obtain a correlation. According to an embodiment of the disclosure, the determining the at least one ambient parameter may comprise determining the at least one ambient parameter based on the correlation.

[0078] According to an embodiment of the disclosure, the identifying at least one IoT device capable of modifying the at least one ambient parameter may comprise determining from a pre-existing map of ambient parameters obtained from the IoT database. According to an embodiment of the disclosure, the identifying at least one IoT device capable of modifying the at least one ambient parameter may comprise determining the primary capabilities along with the secondary capabilities of the IoT devices that can also cause the discontinuity and have an effect on the ambient parameter. According to an embodiment of the disclosure, the identifying at least one IoT device capable of modifying the at least one ambient parameter may comprise determining the location of the IoT devices that are near the VR device that have that capability. According to an embodiment of the disclosure, the identifying at least one IoT device capable of modifying the at least one ambient parameter may comprise checking if the IoT devices can be controlled by the user and don't have restrictions. According to an embodiment of the disclosure, the identifying at least one IoT device capable of modifying the at least one ambient parameter may comprise computing a range to match with the ambience needed change from the current state of the IoT device. According to an embodiment of the disclosure, the identifying at least one IoT device capable of modifying the at least one ambient parameter may comprise obtaining all the past routines that have involved the capability. According to an embodiment of the disclosure, the identifying at least one IoT device capable of modifying the at least one ambient parameter may

comprise classifying the side effects if the ambient parameter related to the IoT device is modified. [0079] According to an embodiment of the disclosure, the controlling the one or more operational characteristics of the at least one IoT device may comprise determining a modification in the at least one ambient parameter to achieve an improved working of the at least one operation. According to an embodiment of the disclosure, the controlling the one or more operational characteristics of the at least one IoT device may comprise controlling the one or more operational characteristics of the at least one IoT device to perform the modification based on a capability of the at least one IoT device.

[0080] According to an embodiment of the disclosure, determining the modification in the at least one ambient parameter may comprise determining a difference between output values of the sensor data and a pre-stored sensor data. According to an embodiment of the disclosure, determining the modification in the at least one ambient parameter may comprise determining, based on the difference, the modification in the at least one ambient parameter to fill a computed gap to achieve the improved working of the at least one operation.

[0081] According to an embodiment of the disclosure, determining the modification in the at least one ambient parameter may comprise monitoring one or more operations of the VR device when the at least one ambient parameter is modified with respect to the modification. According to an embodiment of the disclosure, determining the modification in the at least one ambient parameter may comprise determining, based on the monitoring, one or more effects on operation of the VR device due to the modification in the at least one ambient parameter. According to an embodiment of the disclosure, determining the modification in the at least one ambient parameter may comprise determining a change in the at least one IoT device to fix the one or more effects on the operation of the VR device. According to an embodiment of the disclosure, determining the modification in the at least one ambient parameter may comprise notifying, to the user, the change in the at least one IoT device.

[0082] According to an embodiment of the disclosure, to identify the at least one discontinuity in the at least one operation, the one or more processors may be configured to obtain the sensor data associated with the at least one operation. According to an embodiment of the disclosure, to identify the at least one discontinuity in the at least one operation, the one or more processors may be configured to correlate the sensor data with a pre-stored sensor data indicating an optimal working criteria of the at least one operation to obtain a correlation. According to an embodiment of the disclosure, to identify the at least one discontinuity in the at least one operation, the one or more processors may be configured to identify the at least one discontinuity in the at least one operation based on the correlation.

[0083] According to an embodiment of the disclosure, the correlating the sensor data with the pre-stored sensor data may comprise comparing a continuous state of the sensor data with respect to a state of the pre-stored sensor data.

[0084] According to an embodiment of the disclosure, the at least one discontinuity may indicate a degree of discontinuity in a behaviour of the at least one operation.

[0085] According to an embodiment of the disclosure, to determine the at least one ambient parameter, the one or more processors may be configured to obtain, from a plurality of sensors and the at least one IoT device, sensor data and IoT data. According to an embodiment of the disclosure, to determine the at least one ambient parameter, the one or more processors may be configured to identify, based on the obtained sensor data and the IoT data, at least one of a respective sensor and a respective IoT device associated with the at least one discontinuity in the at least one operation. According to an embodiment of the disclosure, to determine the at least one ambient parameter, the one or more processors may be configured to correlate the at least one ambient parameter with the at least one discontinuity in the at least one operation to obtain a correlation. According to an embodiment of the disclosure, to determine the at least one ambient parameter, the one or more processors may be configured to determine the at least one ambient

parameter based on the correlation. According to an embodiment of the disclosure, to control the one or more operational characteristics of the at least one IoT device, the one or more processors may be configured to determine a modification in the at least one ambient parameter to achieve an improved working of the at least one operation. According to an embodiment of the disclosure, to control the one or more operational characteristics of the at least one IoT device, the one or more processors may be configured to control the one or more operational characteristics of the at least one IoT device to perform the modification based on a capability of the at least one IoT device.

[0086] According to an embodiment of the disclosure, to determine the modification in the at least one ambient parameter, the one or more processors may be configured to determine a difference between output values of the sensor data and a pre-stored sensor data. According to an embodiment of the disclosure, to determine the modification in the at least one ambient parameter, the one or more processors may be configured to determine, based on the difference, the modification in the at least one ambient parameter to fill a computed gap to achieve the improved working of the at least one operation.

[0087] According to an embodiment of the disclosure, the one or more processors may be configured to monitor one or more operations of the VR device when the at least one ambient parameter is modified with respect to the modification. According to an embodiment of the disclosure, the one or more processors may be configured to determine, based on the monitoring, one or more effects on operation of the VR device due to the modification in the at least one ambient parameter. According to an embodiment of the disclosure, the one or more processors may be configured to determine a change in the at least one IoT device to fix the one or more effects on the operation of the VR device. According to an embodiment of the disclosure, the one or more processors may be configured to notify, to the user, the change in the at least one IoT device.

[0088] According to an embodiment of the disclosure, the discontinuity may be an interruption in the sensor data.

[0089] According to an embodiment of the disclosure, the method of controlling a Virtual Reality (VR) device in an internet of Things (IoT) environment, the method may include obtain sensor data from the VR device. According to an embodiment of the disclosure, the method of controlling a Virtual Reality (VR) device in an internet of Things (IoT) environment, the method may include detecting an anomaly in at least one operation of the VR device while a user is using the VR device, based on the sensor data. According to an embodiment of the disclosure, the method of controlling a Virtual Reality (VR) device in an internet of Things (IoT) environment, the method may include determining an ambient condition of the VR device associated with the anomaly.

According to an embodiment of the disclosure, the method of controlling a Virtual Reality (VR) device in an internet of Things (IoT) environment, the method may include identifying an IoT device capable of modifying the ambient condition. According to an embodiment of the disclosure, the method of controlling a Virtual Reality (VR) device in an internet of Things (IoT) environment, the method may include instructing the IoT device to adjust operation to modify the ambient condition to reduce the anomaly.

[0090] According to an embodiment of the disclosure, the anomaly may be an interruption in the sensor data.

[0091] According to an embodiment of the disclosure, above apparatus and method may overcome at least some of the above deficiencies.

Claims

1. A method for a Virtual Reality (VR) device in an Internet of Things (IoT) environment, the method comprising: obtaining sensor data from the VR device; identifying at least one discontinuity in at least one operation associated with the VR device while a user is using the VR device, based on the sensor data; determining at least one ambient parameter around the VR device

associated with the at least one discontinuity in the at least one operation; identifying at least one IoT device capable of modifying the at least one ambient parameter; and controlling one or more operational characteristics of the at least one IoT device to modify the at least one ambient parameter around the VR device to reduce the at least one discontinuity in the at least one operation.

2. The method as claimed in claim 1, wherein the identifying the at least one discontinuity in the at least one operation comprises: obtaining the sensor data associated with the at least one operation; correlating the sensor data with a pre-stored sensor data indicating an optimal working criteria of the at least one operation to obtain a correlation; and identifying the at least one discontinuity in the at least one operation based on the correlation.

3. The method as claimed in claim 2, wherein the correlating the sensor data with the pre-stored sensor data comprises comparing a continuous state of the sensor data with respect to a state of the pre-stored sensor data.

4. The method as claimed in claim 2, wherein the at least one discontinuity indicates a degree of discontinuity in a behaviour of the at least one operation.

5. The method as claimed in claim 1, wherein the determining the at least one ambient parameter comprises: obtaining, from a plurality of sensors and the at least one IoT device, the sensor data and IoT data; identifying, based on the obtained sensor data and the IoT data, at least one of a respective sensor and a respective IoT device associated with the at least one discontinuity in the at least one operation; correlating the at least one ambient parameter with the at least one discontinuity in the at least one operation to obtain a correlation; and determining the at least one ambient parameter based on the correlation.

6. The method as claimed in claim 1, wherein the identifying at least one IoT device capable of modifying the at least one ambient parameter, comprises: determining from a pre-existing map of ambient parameters obtained from the IoT database; determining the primary capabilities along with the secondary capabilities of the IoT devices that can also cause the discontinuity and have an effect on the ambient parameter; determining the location of the IoT devices that are near the VR device that have that capability; checking if the IoT devices can be controlled by the user and don't have restrictions; computing a range to match with the ambience needed change from the current state of the IoT device; obtaining all the past routines that have involved the capability; and classifying the side effects if the ambient parameter related to the IoT device is modified.

7. The method as claimed in claim 1, wherein the controlling the one or more operational characteristics of the at least one IoT device comprises: determining a modification in the at least one ambient parameter to achieve an improved working of the at least one operation; and controlling the one or more operational characteristics of the at least one IoT device to perform the modification based on a capability of the at least one IoT device.

8. The method as claimed in claim 7, wherein determining the modification in the at least one ambient parameter comprises: determining a difference between output values of the sensor data and a pre-stored sensor data; and determining, based on the difference, the modification in the at least one ambient parameter to fill a computed gap to achieve the improved working of the at least one operation.

9. The method as claimed in claim 8, further comprising: monitoring one or more operations of the VR device when the at least one ambient parameter is modified with respect to the modification; determining, based on the monitoring, one or more effects on operation of the VR device due to the modification in the at least one ambient parameter; determining a change in the at least one IoT device to fix the one or more effects on the operation of the VR device; and notifying, to the user, the change in the at least one IoT device.

10. An apparatus configured to connect with a Virtual Reality (VR) device in an Internet of Things (IoT) environment, the apparatus comprising: at least one memory storing instructions; one or more processors operatively connected to the at least one memory, and configured to execute the

instructions to: obtain sensor data from the VR device; identify at least one discontinuity in at least one operation associated with the VR device while a user is using the VR device, based on the sensor data; determine at least one ambient parameter around the VR device associated with the at least one discontinuity in the at least one operation; identify at least one IoT device capable of modifying the at least one ambient parameter; and control one or more operational characteristics of the at least one IoT device to modify the at least one ambient parameter around the VR device to reduce the at least one discontinuity in the at least one operation.

11. The apparatus as claimed in claim 10, wherein to identify the at least one discontinuity in the at least one operation, the one or more processors are configured to: obtain the sensor data associated with the at least one operation; correlate the sensor data with a pre-stored sensor data indicating an optimal working criteria of the at least one operation to obtain a correlation; and identify the at least one discontinuity in the at least one operation based on the correlation.

12. The apparatus as claimed in claim 11, wherein the correlating the sensor data with the pre-stored sensor data comprises comparing a continuous state of the sensor data with respect to a state of the pre-stored sensor data.

13. The apparatus as claimed in claim 11, wherein the at least one discontinuity indicates a degree of discontinuity in a behaviour of the at least one operation.

14. The apparatus as claimed in claim 10, wherein to determine the at least one ambient parameter, the one or more processors are configured to: obtain, from a plurality of sensors and the at least one IoT device, the sensor data and IoT data; identify, based on the obtained sensor data and the IoT data, at least one of a respective sensor and a respective IoT device associated with the at least one discontinuity in the at least one operation; correlate the at least one ambient parameter with the at least one discontinuity in the at least one operation to obtain a correlation; and determine the at least one ambient parameter based on the correlation.

15. The apparatus as claimed in claim 10, wherein to control the one or more operational characteristics of the at least one IoT device, the one or more processors are configured to: determine a modification in the at least one ambient parameter to achieve an improved working of the at least one operation; and control the one or more operational characteristics of the at least one IoT device to perform the modification based on a capability of the at least one IoT device.

16. The apparatus as claimed in claim 15, wherein to determine the modification in the at least one ambient parameter, the one or more processors are configured to: determine a difference between output values of the sensor data and a pre-stored sensor data; and determine, based on the difference, the modification in the at least one ambient parameter to fill a computed gap to achieve the improved working of the at least one operation.

17. The apparatus as claimed in claim 16, wherein the one or more processors are configured to: monitor one or more operations of the VR device when the at least one ambient parameter is modified with respect to the modification; determine, based on the monitoring, one or more effects on operation of the VR device due to the modification in the at least one ambient parameter; determine a change in the at least one IoT device to fix the one or more effects on the operation of the VR device; and notify, to the user, the change in the at least one IoT device.

18. The method of claim 1, wherein the discontinuity is an interruption in the sensor data.

19. The apparatus of claim 9, wherein the discontinuity is an interruption in the sensor data.

20. A non-transitory computer-readable storage medium storing instructions that, when executed by at least one processor, cause the at least one processor to: obtain sensor data from the VR device; identify at least one discontinuity in at least one operation associated with the VR device while a user is using the VR device, based on the sensor data; determine at least one ambient parameter around the VR device associated with the at least one discontinuity in the at least one operation; identify at least one IoT device capable of modifying the at least one ambient parameter; and control one or more operational characteristics of the at least one IoT device to modify the at least

one ambient parameter around the VR device to reduce the at least one discontinuity in the at least one operation.
